■ CarePoint Hospital Management System

Relational Database Schema Technical Summary & Specification Document

Target Database: MySQL 8.0 Enterprise • **Tables:** 13 Verified Relations • **Normalization:** 3NF / BCNF

This document provides the definitive data dictionary, relational table specifications, primary/foreign key definitions, and referential constraints extracted from the implemented Hospital Management System codebase. All definitions reflect verified MySQL DDL structures and Java/Node.js backend models.

1. Relational Table Specifications & Data Dictionary

Table: Specializations (Doctor Management)

Column Name	Data Type	Key Type	Nullability & Constraints
specialization_id	INT	**PK**	PK, AUTO_INCREMENT
specialization_name	VARCHAR(50)	-	UNIQUE, NOT NULL

Table: Doctor (Doctor Management)

Column Name	Data Type	Key Type	Nullability & Constraints
doctor_id	INT	**PK**	PK, AUTO_INCREMENT
doctor_name	VARCHAR(50)	-	NOT NULL
specialization_id	INT	**FK**	FK -> Specializations
qualification	VARCHAR(50)	-	NOT NULL
consultation_fee	DECIMAL(10,2)	-	DEFAULT 500.00
contact	VARCHAR(15)	-	NOT NULL

Table: Doctor_Schedule (Doctor Management)

Column Name	Data Type	Key Type	Nullability & Constraints
schedule_id	INT	**PK**	PK, AUTO_INCREMENT
doctor_id	INT	**FK**	FK -> Doctor, NOT NULL
day_of_week	VARCHAR(15)	-	NOT NULL
start_time	TIME	-	NOT NULL
end_time	TIME	-	NOT NULL

Table: Patients (Patient & Outpatient Core)

Column Name	Data Type	Key Type	Nullability & Constraints
patient_id	INT	**PK**	PK, AUTO_INCREMENT
patient_name	VARCHAR(50)	-	NOT NULL
gender	VARCHAR(10)	-	Male/Female/Other
age	INT	-	CHECK >= 0
blood_group	VARCHAR(5)	-	NOT NULL
contact	VARCHAR(15)	-	INDEXED, NOT NULL
address	VARCHAR(100)	-	Nullable
registration_date	DATE	-	NOT NULL

Table: Appointments (Patient & Outpatient Core)

Column Name	Data Type	Key Type	Nullability & Constraints
appointment_id	INT	**PK**	PK, AUTO_INCREMENT
patient_id	INT	**FK**	FK -> Patients, NOT NULL
doctor_id	INT	**FK**	FK -> Doctor, NOT NULL
appointment_date	DATE	-	INDEXED, NOT NULL
appointment_time	TIME	-	NOT NULL
room_number	VARCHAR(10)	-	NOT NULL
status	VARCHAR(20)	-	DEFAULT 'Scheduled'

Table: Emergency (Patient & Outpatient Core)

Column Name	Data Type	Key Type	Nullability & Constraints
emergency_id	INT	**PK**	PK, AUTO_INCREMENT
patient_id	INT	**FK**	FK -> Patients (Inferred)
emergency_type	VARCHAR(100)	-	NOT NULL
priority_level	VARCHAR(20)	-	Critical/High/Med/Low
status	VARCHAR(30)	-	DEFAULT 'Admitted'
assigned_doctor	INT	**FK**	FK -> Doctor (Inferred)
arrival_date	DATE	-	NOT NULL
arrival_time	TIME	-	NOT NULL

Table: Prescriptions (Clinical & Financial Encounters)

Column Name	Data Type	Key Type	Nullability & Constraints
prescription_id	INT	**PK**	PK, AUTO_INCREMENT
appointment_id	INT	**FK**	FK -> Appointments, UNIQUE
diagnosis	VARCHAR(100)	-	NOT NULL
medicine	VARCHAR(100)	-	NOT NULL
next_visit_date	DATE	-	Nullable
remarks	VARCHAR(100)	-	Nullable

Table: Billing (Clinical & Financial Encounters)

Column Name	Data Type	Key Type	Nullability & Constraints
bill_id	INT	**PK**	PK, AUTO_INCREMENT
appointment_id	INT	**FK**	FK -> Appointments, UNIQUE
amount	DECIMAL(10,2)	-	CHECK >= 0
bill_date	DATE	-	NOT NULL
payment_method	VARCHAR(20)	-	UPI/Card/Cash
payment_status	VARCHAR(20)	-	INDEXED (Paid/Pending)

Table: Lab_Tests (Clinical & Financial Encounters)

Column Name	Data Type	Key Type	Nullability & Constraints
test_id	INT	**PK**	PK, AUTO_INCREMENT
patient_id	INT	**FK**	FK -> Patients, NOT NULL
doctor_id	INT	**FK**	FK -> Doctor, NOT NULL
test_name	VARCHAR(100)	-	NOT NULL
test_date	DATE	-	NOT NULL
cost	DECIMAL(10,2)	-	CHECK >= 0
result	VARCHAR(255)	-	Default 'Pending Analysis'
status	VARCHAR(30)	-	DEFAULT 'Pending'

Table: Bed_Allocation (Facility, Security & Audit)

Column Name	Data Type	Key Type	Nullability & Constraints
allocation_id	INT	**PK**	PK, AUTO_INCREMENT
patient_id	INT	**FK**	FK -> Patients, NOT NULL
ward_type	VARCHAR(50)	-	ICU/General/Private AC
bed_number	VARCHAR(10)	-	NOT NULL
admit_date	DATE	-	NOT NULL
discharge_date	DATE	-	Nullable
daily_charge	DECIMAL(10,2)	-	NOT NULL
status	VARCHAR(20)	-	DEFAULT 'Occupied'

Table: Feedback (Facility, Security & Audit)

Column Name	Data Type	Key Type	Nullability & Constraints
feedback_id	INT	**PK**	PK, AUTO_INCREMENT
patient_id	INT	**FK**	FK -> Patients, NOT NULL
rating	INT	-	CHECK (1 TO 5)
feedback_date	DATE	-	NOT NULL
comments	VARCHAR(100)	-	Nullable

Table: Users (Facility, Security & Audit)

Column Name	Data Type	Key Type	Nullability & Constraints
user_id	INT	**PK**	PK, AUTO_INCREMENT
username	VARCHAR(50)	-	UNIQUE, NOT NULL
password	VARCHAR(128)	-	SHA-256 Digest
full_name	VARCHAR(100)	-	NOT NULL
role	VARCHAR(20)	-	Admin/Doctor/Recep/Patient
created_at	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP

Table: Audit_Logs (Facility, Security & Audit)

Column Name	Data Type	Key Type	Nullability & Constraints
log_id	INT	**PK**	PK, AUTO_INCREMENT
action_type	VARCHAR(50)	-	NOT NULL
record_id	INT	-	Logical FK (Entity ID)
log_time	TIMESTAMP	-	DEFAULT CURRENT_TIMESTAMP
details	TEXT	-	Nullable

2. Referential Integrity & Relationship Mapping

The table below details all 13 entity relationships, distinguishing between database engine enforced foreign keys and application-enforced logical relationships:

Parent Entity	Child Entity	Card.	Relationship Label	Enforcement Evidence & SQL Constraint
Specializations	**Doctor**	1:N	classifies	**Enforced FK** Explicit FK: doctor.specialization_id -> specializations.specialization_id
Doctor	**Doctor_Schedule**	1:N	works_shifts	**Enforced FK** Explicit FK: doctor_schedule.doctor_id -> doctor.doctor_id
Doctor	**Appointments**	1:N	attends	**Enforced FK** Explicit FK: appointments.doctor_id -> doctor.doctor_id
Patients	**Appointments**	1:N	books	**Enforced FK** Explicit FK: appointments.patient_id -> patients.patient_id
Appointments	**Prescriptions**	1:1	issues	**Enforced FK** Explicit FK: prescriptions.appointment_id -> appointments.appointment_id (Unique per consult)
Appointments	**Billing**	1:1	generates	**Enforced FK** Explicit FK: billing.appointment_id -> appointments.appointment_id (Unique per consult)
Patients	**Bed_Allocation**	1:N	allocated_bed	**Enforced FK** Explicit FK: bed_allocation.patient_id -> patients.patient_id
Patients	**Feedback**	1:N	submits	**Enforced FK** Explicit FK: feedback.patient_id -> patients.patient_id
Patients	**Lab_Tests**	1:N	undergoes	**Enforced FK** Explicit FK: lab_tests.patient_id -> patients.patient_id (ON DELETE CASCADE)
Doctor	**Lab_Tests**	1:N	orders	**Enforced FK** Explicit FK: lab_tests.doctor_id -> doctor.doctor_id (ON DELETE CASCADE)
Patients	**Emergency**	1:N	admitted_as	**Inferred (App Logic)** Inferred: emergency.patient_id = patients.patient_id enforced via DAO and API queries
Doctor	**Emergency**	1:N	assigned_doctor	**Inferred (App Logic)** Inferred: emergency.assigned_doctor = doctor.doctor_id enforced via backend JOIN queries
Appointments	**Audit_Logs**	1:N	audits	**Inferred (App Logic)** Inferred: audit_logs.record_id = appointments.appointment_id logged on trigger trg_AuditAppointmentCancel

Verification Certification: Extracted directly from live MySQL 8.0 schema and repository source code. Repository: https://github.com/nisargjoshi2006-web/HospitalManagementSystem